

WHY TRANSFORMATION?

A Strategy Can Be Correct and Still Fail

Strategy defines direction. It does not, by itself, define the organizational system **capable** of executing that direction — which is the entire subject of this dossier.

STRATEGY, TRANSFORMATION, EXECUTION — NOT SYNONYMS

Strategy is what the organization intends to achieve and where it chooses to compete or create value. **Transformation** is a shift in the organization's underlying operating logic — not merely its activity level. **Execution** is the set of mechanisms — process, coordination, communication — through which intended action actually occurs. **PRIMARY SOURCE** Porter's foundational distinction separates operational effectiveness ("performing similar activities better than rivals") from strategic positioning ("performing different activities... or performing similar activities in different ways").¹ Chandler's historical thesis — structure follows strategy — established that a chosen strategy still requires a compatible organizational system to be realized.² **HISTORICAL**

TRANSFORMATION VS. CONTINUOUS IMPROVEMENT

No universal taxonomy separates these two terms. The closest rigorous anchor is Bartunek & Moch's academic distinction between orders of organizational change³ — change *within* an existing schema, versus change *to* the schema itself, versus building the capacity to repeatedly change schemas. D4 reads continuous improvement as closer to the first order, and transformation as involving the second and third — an interpretation, not the authors' own claim.

First-Order Change	Second-Order Change	Third-Order Change
Bartunek & Moch, in their own terms: change within an existing organizational schema.	Change to the underlying schema itself — the frame of reference shifts.	Building the organizational capacity to repeatedly examine and change schemas.

ANALYTICAL INTERPRETATION D4 reads continuous improvement as operating within an existing schema, and transformation as reaching into the second/third order — a D4 application, not Bartunek & Moch's own vocabulary.

Strategy defines direction.

Operating model defines how the organization executes that direction.

Transformation is the process of moving from one state to the other.

ANALYTICAL SYNTHESIS The final formulation this dossier builds toward — introduced here, earned by Page 20.

Strategy defines direction. Operating model defines how the organization executes that direction.

WHY STRATEGY FAILS TO BECOME EXECUTION

Kaplan & Norton document that organizations routinely keep planning, budgeting and reviewing performance exactly as before, even after adopting a new strategy — a fragmentation across separate management processes, not a single failure of will.⁴ Sull, Homkes & Sull find middle managers frequently unable to name their own company's top-five priorities.⁵

PROJECT OUTPUT ≠ CAPABILITY ≠ TRANSFORMATION

A completed project is not a completed transformation. Prahalad & Hamel distinguish an organization's visible outputs from the durable, collective capability beneath them⁶ — the distinction this dossier returns to on Page 4.

01–05 Foundations Why transform, the execution gap, the transformation thesis	06–10 Operating Model Dimensions, target state, trade-offs, governance, process	11–15 Enablement Data, performance, capability, incentives, change	16–19 Execution Portfolio, roadmap, control, embedding	20 Synthesis The complete transformation operating model
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¹ Porter, "What Is Strategy?", HBR 1996
² Chandler, Strategy and Structure, 1962
³ Bartunek & Moch, JABS 1987

⁴ Kaplan & Norton, HBS WP 05-071
⁵ Sull, Homkes & Sull, HBR 2015
⁶ Prahalad & Hamel, HBR 1990

Why a Sophisticated Strategy Still Fails

Four separately evidenced failure clusters — not one unified academic taxonomy — explain the **gap** between what an organization says and what it does.



FOUR EVIDENCE-BACKED FAILURE CLUSTERS

1. Priority & Communication

Sull, Homkes & Sull: managers report cross-unit coordination — not vertical alignment — as the real barrier; many cannot name their own top-five priorities.¹ The mechanism: without a short, shared, ranked list, competing local priorities each get pursued in good faith, and coordination collapses at the seams between units rather than at the top of the hierarchy.

2. Process, Budget & Incentive Fragmentation

Kaplan & Norton: planning, budgeting, HR review and performance reporting routinely run as disconnected processes.² The mechanism: a strategy can be formally adopted while the calendar of processes that actually allocate money, people and attention keeps running on its prior, un-updated logic.

3. Decision-Rights Failure

Rogers & Blenko: accountability assigned without matching decision authority stalls decisions.³

INSTITUTIONAL The mechanism: someone is held responsible for an outcome without the authority to make the calls that determine it, so decisions escalate, stall, or get re-litigated.

4. Resource Allocation

Bower: funded initiatives — not the strategy document — reveal an organization's de facto priorities.⁴ The mechanism: capital and staffing decisions are made proposal-by-proposal, by different people, on different timelines than the strategy statement — so the aggregate pattern of funding can diverge from stated intent without anyone deciding it should.

EVIDENCE GAP

A ten-factor failure list (capability, data, leadership alignment, etc.) is sometimes proposed in practice. Only the four clusters above are independently, separately evidenced in this research; the remainder should not be presented as an established diagnostic framework.

READING THE GAP DIAGRAM

"Formal strategy" is what leadership states and publishes. "Actual organizational behaviour" is what the calendar, the budget and the org chart actually reward and permit. The gap between them is not a single failure but the sum of the four clusters at left — which is why closing it requires four separate diagnostic questions, not one motivational fix.

GALBRAITH'S STAR MODEL

INSTITUTIONAL · DESIGN FRAMEWORK

Five interdependent levers — Strategy, Structure, Processes, Rewards, People — offered as a way of thinking about alignment.⁵ Galbraith's own claim is a design principle: change one lever without the others and the organization pulls itself out of alignment. It supports the design argument on this page; it is not itself causal evidence that misalignment across these five variables produces the four failure clusters above, which come from separate empirical studies.

DE FACTO PRIORITIES

Bower's field research: the initiatives that actually receive capital and management attention are the clearest evidence of an organization's real strategy — independent of what the strategy document says.⁴ This is the resource-allocation cluster in its original form; Page 14 develops it further into incentive design.

KEY TAKEAWAY

The gap between formal strategy and actual behaviour is diagnosable — through priorities, process fragmentation, decision rights, and resource allocation — not merely a motivational shortfall.

¹ Sull, Homkes & Sull, HBR 2015

² Kaplan & Norton, HBS WP 05-071

³ Rogers & Blenko, HBR 2006

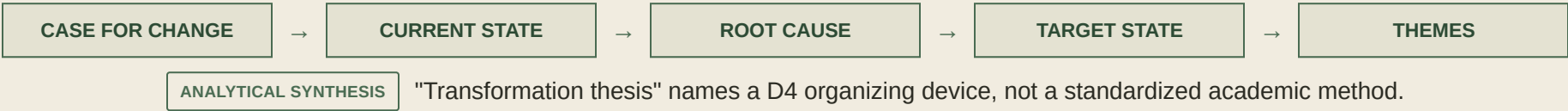
⁴ Bower, HBS 1970

⁵ Galbraith, Star Model

THE TRANSFORMATION THESIS

What Exactly Are We Trying to Change?

A transformation thesis is a **diagnosis** and an integrating logic — not a list of projects assembled before anyone has asked what actually needs to change.



CASE FOR CHANGE

Kotter's first of eight documented transformation errors is insufficient shared urgency — his own observation, not an independently replicated statistic.¹

HISTORICAL

 Kotter uses "transformation" and "change" largely interchangeably, rather than as technically distinct terms. A case for change states, concretely, what happens if the organization does not change — the cost of the status quo — not merely that change is generally desirable.

CURRENT STATE, HONESTLY DESCRIBED

A current-state description that only restates the case for change (i.e. only what is wrong) is incomplete: it must also capture what already works, since a target state that discards functioning capability alongside broken capability creates unnecessary transition cost and unnecessary resistance (Page 15).

ROOT CAUSE VS. SYMPTOM

Senge's "shifting the burden" archetype: a **symptomatic solution** relieves symptoms without resolving the underlying cause, so symptoms return; a **fundamental solution** addresses the cause directly — and relieving symptoms quickly reduces an organization's motivation to pursue the fundamental fix.²

PRIMARY SOURCE

 Applied to D4: a dashboard purchased to fix a "we lack visibility" symptom, without fixing the decision-rights ambiguity (Page 9) that made the information unusable in the first place, is a symptomatic solution — it will need to be repeated.

THESIS, NOT AN INITIATIVE LIST

WEAKER — INITIATIVE LIST

"Implement dashboard. Hire an analyst. Create a committee. Update the procurement process."

ANALYTICAL ILLUSTRATION

Constructed contrast, not a documented case.

STRONGER — THESIS

"The organization requires an integrated decision and execution model that embeds the priority into planning, operations, accountability, information and resource allocation."

TARGET STATE

Not structure alone: strategic outcomes, capabilities, operating model, responsibilities, processes, systems and performance requirements together.

TRANSFORMATION VALUE LOGIC

Change → capability → business outcome. Prahalad & Hamel: a built capability generates outcomes across multiple products and markets, not one initiative producing one result.³

D1/D2/D3 FIREWALL

This page diagnoses organizational change requirements only. Financial valuation (D1), impact materiality (D2) and scenario analysis (D3) are referenced elsewhere in the four-dossier system, not re-derived here.

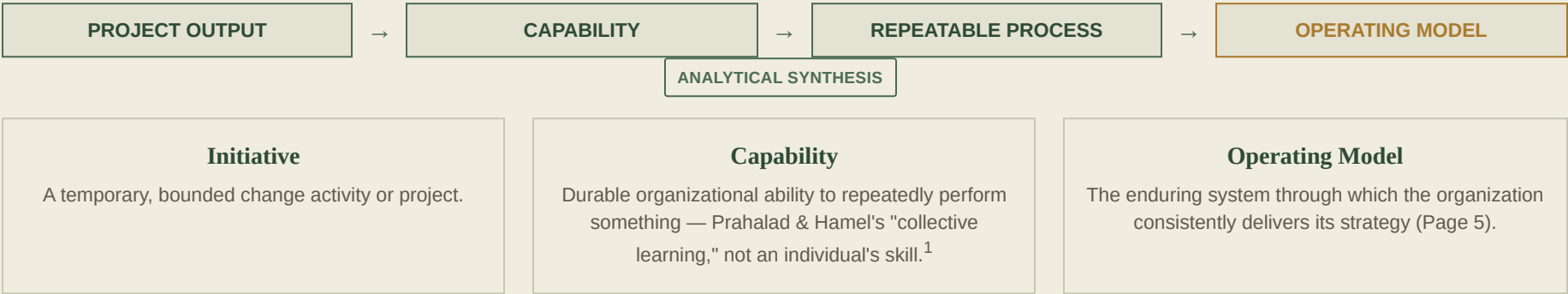
1 Kotter, HBR 1995
2 Senge, The Fifth Discipline, 1990

3 Prahalad & Hamel, HBR 1990

FROM INITIATIVES TO OPERATING MODEL

Why a Project List Is Not a Transformation

A collection of completed projects does not, by itself, produce a durable **capability** or an operating model.



TEMPORARY PROGRAM VS. ENDURING MODEL

Kaplan & Norton propose a small, *permanent* function (six to eight people, reporting to the CEO) precisely because treating strategy execution as a temporary program — with no permanent home — is itself a documented cause of process fragmentation.² This is their specific proposed structure, not a universal prescription that every organization needs an equivalent office. The underlying principle generalizes: whatever coordinates strategy execution needs to outlive any single initiative, or it re-fragments the moment that initiative's team disbands — the same argument Page 19 makes about embedding.

THE "70% FAIL" STATISTIC — INVESTIGATED, NOT REPEATED

The figure is traced to Hammer & Champy's own "unscientific estimate" (1993), offered without a study behind it, and then to Beer & Nohria's 2000 HBR restatement "with no reference or evidence whatsoever."³ From there it was repeated by successive commentators until it acquired the appearance of an established finding — a citation chain with no empirical study at its origin.

ACADEMIC EVIDENCE Hughes (2011), *Journal of Change Management*: "there is no valid and reliable empirical evidence to support" the 70% narrative — a direct academic rejection of the figure's evidentiary basis.⁴

PRIMARY SOURCE IBM (2008), 1,532 practitioners across 15 nations: 41% of projects met objectives; top-quartile organizations reached 80% success versus 8% for bottom-quartile — variation by organizational capability, not a fixed universal failure rate.⁵ This is the more useful finding: success correlates with specific, buildable capabilities, not with some fixed law of transformation.

STANDING CONCLUSION

The 70% figure should not be treated as a reliable benchmark or repeated as fact. No replacement percentage is substituted in its place — the honest conclusion is that success rates vary by organizational capability, not that a different fixed number is correct.

KEY TAKEAWAY

A capability is not established because one employee knows something, one project delivered something, or one system exists — it requires repeatability, ownership and process integration.

¹ Prahalad & Hamel, HBR 1990

² Kaplan & Norton, HBS WP 05-071

³ Hammer & Champy, 1993

⁴ Hughes, Journal of Change Management 2011

⁵ IBM, Making Change Work, 2008

WHAT IS AN OPERATING MODEL?

How Strategy Becomes a Repeatable System

An operating model is broader than an org chart, narrower than a business model, and more than any single **process**.



ANALYTICAL SYNTHESIS This relationship architecture, not a nested-containment claim.

Process ≠ operating model. A process is one flow of work; the operating model is the broader system of structure, governance, capability, data and performance measurement that governs many processes at once. **Structure ≠ operating model.** Structure — reporting lines and hierarchy — is one component within it, not a synonym for it. **Business model ≠ operating model.** The business model states the logic of value creation; the operating model is the configuration that carries out that logic day to day — two organizations can share a business model (e.g. subscription software) while running materially different operating models (centralized vs. federated support, in-house vs. outsourced delivery).

ANCHOR DEFINITION

ACADEMIC EVIDENCE MIT CISR (Ross, Weill & Robertson): an operating model is "the necessary level of business process integration and standardization for delivering goods and services to customers" — a specific, two-dimensional construct (integration and standardization), not a vague catch-all.¹ Their framework treats these two dimensions as choices an organization makes deliberately, not defaults that emerge on their own.

OPERATING MODEL VS. BUSINESS MODEL

Osterwalder & Pigneur: a business model describes "the rationale of how an organization creates, delivers, and captures value."² The operating model is the system that executes that logic — a subset of business-model delivery, not a synonym. Changing a business model (e.g. shifting from product sale to subscription) typically forces operating-model change; the reverse is not automatic — an operating model can be redesigned without changing the underlying value-creation logic at all.

OPERATING MODEL VS. ORGANIZATIONAL STRUCTURE

An org chart answers "who reports to whom." The operating model additionally answers who decides what, which processes exist, what data and systems support them, and how performance is measured — an org chart redraw with none of the rest changed is a structure change, not an operating-model change.

D4 EXECUTION EXAMPLE

Strategy: integrate a priority into capital planning. Operating-model implication: decision rights, data, the planning process, capability, KPIs, governance — execution mechanics, not the scenario analysis behind the priority itself (D3's domain).

OPERATING MODEL DIMENSIONS

What Makes an Operating Model Work?

No single source establishes one universal set of dimensions. Three frameworks **converge** on a smaller core than any one of them claims alone.

FRAMEWORK	STATUS	ELEMENTS
Galbraith Star Model	DESIGN FRAMEWORK	Strategy, Structure, Processes, Rewards, People — "all policies must be aligned with one another." ¹
McKinsey 7-S	CONSULTING FRAMEWORK	Strategy, Structure, Systems (hard) + Shared Values, Skills, Staff, Style (soft). Waterman, Peters & Phillips, <i>Business Horizons</i> , 1980: "structure is not organization." ²
MIT CISR	ACADEMIC EVIDENCE	Business-process integration and standardization requirements — a narrower, more precise construct (Page 5). ³
D4 Synthesis (8 dimensions)	ANALYTICAL SYNTHESIS	Strategy/Direction · Structure · Governance/Decision Rights · Processes · People/Capabilities · Data/Technology · Performance Management · Incentives/Rewards

What is consistently present across all three: strategy/direction as the anchor, structure, some form of process, some form of people/capability. What differs: an explicit rewards lever (Galbraith, 7-S) versus an implicit one (CISR); data/technology and governance as standalone dimensions appear only in later, more contemporary treatments. **The eight-dimension D4 list is a synthesis assembled from this convergence — not a documented consensus.**

THE OPERATING MODEL IS A SYSTEM, NOT A CHECKLIST

ACADEMIC EVIDENCE Lawrence & Lorsch's field study of the plastics industry found high-performing organizations pairing high differentiation (units suited to distinct sub-environments) with correspondingly strong integration mechanisms — misalignment between the two "predictably creates friction and delays."⁴ This is empirical evidence for interdependence, not merely an assertion: it means the eight dimensions above cannot be improved one at a time in isolation and expected to sum to a better operating model. A dimension strengthened without its interdependent partners can make overall performance worse, not better — a stronger process without a matching decision-rights fix simply routes ambiguity through a faster pipe.

MISALIGNMENT — ANALYTICAL ILLUSTRATION

Strong strategy + weak incentives = execution contradiction, because people are asked to pursue one goal while being paid for another. Strong technology + weak process = automation of a bad process, delivering the same wrong outcome faster. Clear accountability + no decision authority = accountability without control, the exact failure mode Rogers & Blenko document on Page 9. Rich data + no performance-review cadence = information no one acts on.

¹ Galbraith, Star Model

² Waterman, Peters & Phillips, Business Horizons 1980

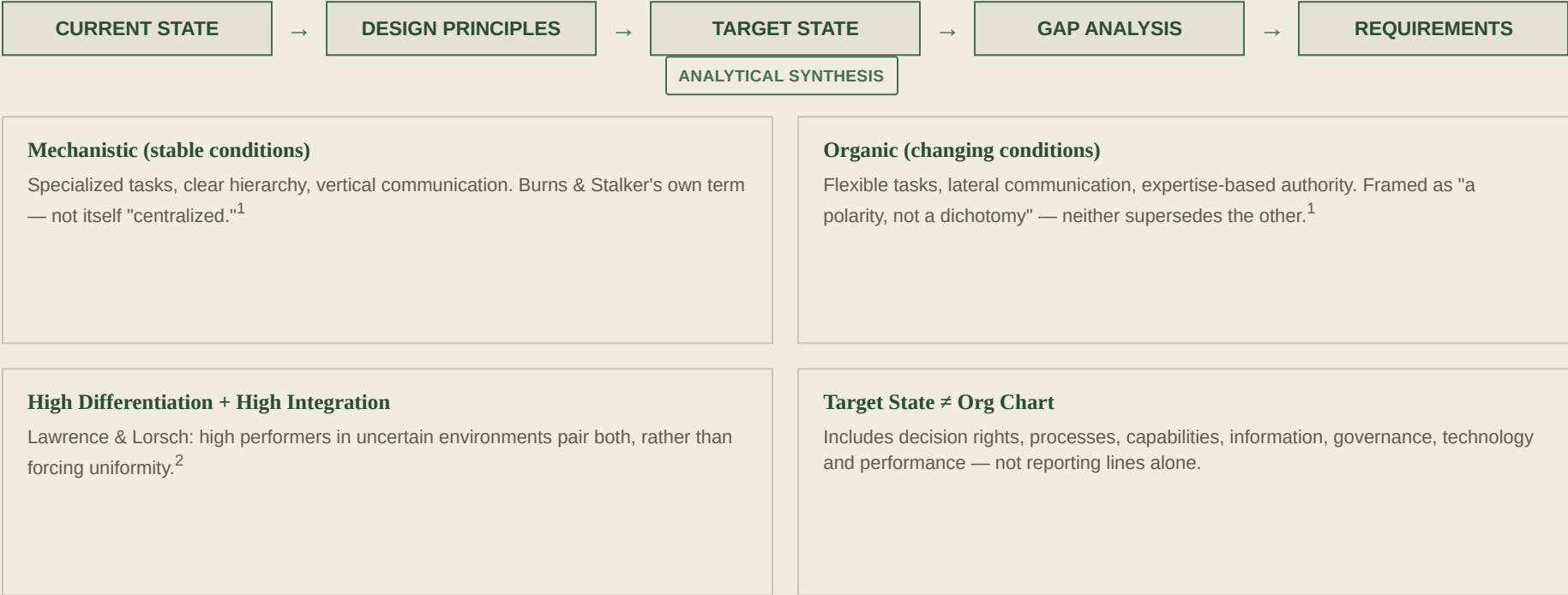
³ Ross, Weill & Robertson, 2006

⁴ Lawrence & Lorsch, 1967

TARGET OPERATING MODEL

What Should the Organization Look Like?

Centralization and standardization are context-dependent **trade-offs**, not settled questions with one correct answer.



ANALYTICAL INTERPRETATION

Mapping Burns & Stalker's mechanistic/organic terms onto "centralized/decentralized" is a D4 reading, not their own vocabulary. Mechanistic is not "bad"; organic is not "good" — Burns & Stalker frame the pair explicitly as fitting different rates of environmental change, not as a maturity ladder from worse to better.

Centralization concentrates decision rights and standard-setting at the center; decentralization pushes them toward the units closest to the customer or operation. Neither is a default answer: centralization buys consistency and scale economics at the cost of local responsiveness, while decentralization buys contextual fit at the cost of duplicated effort and inconsistent customer experience. The same trade-off recurs, in sharper form, on Page 8.

DESIGN PRINCIPLES

Standardize where consistency and scale create value; preserve local discretion where context materially affects outcomes.

ANALYTICAL ILLUSTRATION

A useful test: if two units solving the same problem differently creates confusion or risk (e.g. two different data-security standards), standardize; if it reflects a genuine difference in customer or market context (e.g. two different sales motions for two different customer segments), preserve discretion.

SPECIALIST/COE VS. EMBEDDED; GLOBAL VS. LOCAL

A center of excellence concentrates specialist capability centrally and serves the organization on demand — it protects scarce expertise and consistency, but can become a bottleneck if demand outstrips its capacity. An embedded model places that same capability inside business units — faster access and better contextual fit, at the cost of inconsistent practice and duplicated headcount across units. Global vs. local decision rights follow the identical logic applied to geography rather than function. These are extensions of the same contingency logic above, not independently evidenced beyond it.

ANALYTICAL SYNTHESIS

GAP ANALYSIS, NOT ASPIRATION

A target-state design is only useful once it is compared against the honestly-described current state (Page 3) to produce a specific list of requirements — the design principles above answer "what should we optimize for," gap analysis answers "what specifically has to change to get there."

¹ Burns & Stalker, The Management of Innovation, 1961

² Lawrence & Lorsch, Organization and Environment, 1967

OPERATING-MODEL DESIGN TRADE-OFFS

What Should the Operating Model Optimize For?

Contingency theory: no single best organizational design exists independent of **context** — but not every trade-off below is equally evidenced.

DESIGN CHOICE	ADVANTAGE	COST / RISK	EVIDENCE STATUS
Central control	Consistency, scale, specialization	Slower local decisions, bottlenecks	EVIDENCED
Local flexibility	Responsiveness, contextual fit	Duplication, inconsistency	EVIDENCED
Standardization	Control, efficiency, comparability	Reduced adaptability	EVIDENCED
Customization	Local fit	Complexity	EVIDENCED
Efficiency ↔ Resilience	Lean vs. redundant capacity	Fragility vs. cost	ANALYTICAL EXTENSION
Speed ↔ Control	Fast decisions vs. oversight	Risk vs. bottlenecks	ANALYTICAL EXTENSION
Specialization ↔ Integration	Expertise vs. coordination	Silos vs. complexity	ANALYTICAL EXTENSION

The first four rows extend directly from Burns & Stalker (1961) and Lawrence & Lorsch (1967), both already established on Page 7. The remaining three are logically consistent extensions of the same contingency logic, not independently, separately evidenced in this research — and should be presented as such rather than as six equally-weighted academic dichotomies. This distinction matters in practice: a design team can cite genuine field evidence for the first four rows, but should present the last three as reasoned inference, not settled science.

ACADEMIC EVIDENCE Donaldson's structural contingency theory formalizes "fit" as the central construct: misfit between structure and context predicts performance decline; organizations tend to adapt back toward fit over time — a directional tendency, not a deterministic law.¹ "Fit" here means a specific pairing — a given environment (stable or turbulent, simple or complex) has a corresponding structure that performs best in it; there is no single structure that performs best across all environments.

There is no single best operating-model design independent of context.

Practically, this means the trade-off table above is a diagnostic instrument, not a scorecard to be maximized on every row simultaneously. An organization in a stable, predictable environment can rationally choose more central control and standardization; one facing volatile, fragmented markets can rationally choose more local flexibility and customization — the same design choice can be correct for one business and wrong for another, and wrong for the same business if its environment later changes.

¹ Donaldson, The Contingency Theory of Organizations, 2001

² Burns & Stalker, 1961

³ Lawrence & Lorsch, 1967

GOVERNANCE & DECISION RIGHTS

Who Actually Has Authority to Decide?

Accountability without **authority** produces dysfunctional execution — and a role-involvement chart does not, by itself, resolve who decides.

ACCOUNTABILITY ≠ AUTHORITY

INSTITUTIONAL GUIDANCE

Rogers & Blenko: "ambiguity over who's accountable for which decisions" stalls organizations even when respected companies are involved.¹ Their case illustration — marketers and product developers both assuming responsibility for the same decision — shows how accountability without matching authority produces paralysis, not merely delay.

GOVERNANCE, MANAGEMENT, OVERSIGHT

ANALYTICAL SYNTHESIS

Governance = the mechanisms through which decisions and accountability are structured. Management = the ongoing exercise of authority to direct resources. Oversight = monitoring whether decisions and outcomes conform to expectations, without necessarily holding primary decision authority. No single source cleanly distinguishes all three in an execution (non-board-governance) sense.

RACI — WORK INVOLVEMENT

Responsible, Accountable, Consulted, Informed.

INSTITUTIONAL GUIDANCE

PMI: clarifies stakeholder roles and serves as the baseline of a communications plan.² PMI separates the RACI tool from leadership itself — "there must be one leader that presides over the project team." RACI answers who does the work, who signs off, who is asked for input, and who merely needs to know — a work-involvement map. RACI does not, by itself, resolve contested decision authority: two people can both be marked "Accountable" on adjacent charts and the tool alone will not adjudicate the conflict.

ANALYTICAL SYNTHESIS

RAPID — DECISION ROLES

Recommend, Agree, Perform, Input, Decide.

CONSULTING FRAMEWORK

Bain & Company (Rogers & Blenko, 2006): answers specifically "who plays what role in this decision" — not general task involvement.¹ Critically, RAPID names exactly one "D" (Decide) role per decision — its entire design purpose is to prevent the accountability-without-authority failure this page opens with, by forcing a single named decider before the decision is made, not after it stalls.

DECISION-LEVEL DESIGN

Enterprise standards, risk appetite and major technology architecture typically warrant central decision rights; customer-facing and context-dependent operational decisions typically warrant local ones. **ANALYTICAL ILLUSTRATION** The underlying test is the same contingency logic from Pages 7–8: centralize the decision where consistency materially matters, localize it where context materially matters.

GOVERNANCE FORUMS & ESCALATION

A forum should exist because it enables a decision, coordination or accountability mechanism — not because governance "looks complete." Each recurring forum should be traceable to a specific decision type it exists to make; a forum that exists but makes no decisions is a reporting meeting, not governance. Escalation requires a clear trigger (what condition sends an issue upward), a destination (which named forum or role receives it) and authority (that destination can actually resolve it) — an escalation path missing any one of the three produces delay without resolution.

The two tools answer different questions and are not substitutes: RACI is suited to mapping who is involved in delivering a piece of work; RAPID (or an equivalent decision-rights tool) is suited to resolving who has final say over a specific decision. Using RACI's "Accountable" role as if it settled decision authority is a common, avoidable confusion between the two.

1 Rogers & Blenko, HBR 2006 / Bain & Company

2 PMI, Roles, Responsibilities and Resources

PROCESS INTEGRATION

Where Does Strategy Enter the Way the Company Works?

A priority is not embedded because a policy exists. It must enter recurring processes — owned **end-to-end**, not by any one function alone.



PRIMARY SOURCE Hammer: "the emergent role of the cross-functional process owner, responsible for the design and performance, but not the people" — explicitly distinct from a functional manager, who owns both people and function.¹ As work becomes more broadly scoped and team-based across boundaries, a single accountable owner for the entire workflow prevents each function optimizing its own slice at the expense of the end-to-end outcome. The functional manager still owns headcount, budget and career development within their function; the process owner owns whether the workflow that crosses those functions actually works end to end — the two roles are complementary, not competing, but only if their respective authority is made explicit.

Process mapping makes this visible: documenting a workflow as it actually crosses Finance, Operations, Procurement, Risk and Technology — rather than as each function separately believes it to work — routinely surfaces handoffs no single function was accountable for. Standardization and controls then apply to the mapped end-to-end process, not to each function's internal step in isolation, which is what keeps a control from being satisfied locally while the overall process still fails.



PROCESS TAXONOMY

INSTITUTIONAL APQC's Process Classification Framework (1992) is described by APQC as the world's most widely used cross-industry business-process taxonomy — used for benchmarking, definition and content management.² A shared taxonomy gives functions a common name for "the same process" before anyone tries to map or own it — without it, Finance and Operations can each believe they already have a functioning process while describing two different things.

HANDOFFS

Risk: information loss, delay, duplication, inconsistent standards. Not every handoff is bad — some are necessary; the risk rises with functional fragmentation and no end-to-end owner. Each handoff is a point where the receiving function can silently apply its own standard rather than the sending function's — which is exactly the failure a single process owner exists to prevent.

INTEGRATION FAILURE — ANALYTICAL ILLUSTRATION

Policy without process = intent without execution. Process without data = manual workaround. Process without owner = coordination gap. Process without performance review = no feedback loop.

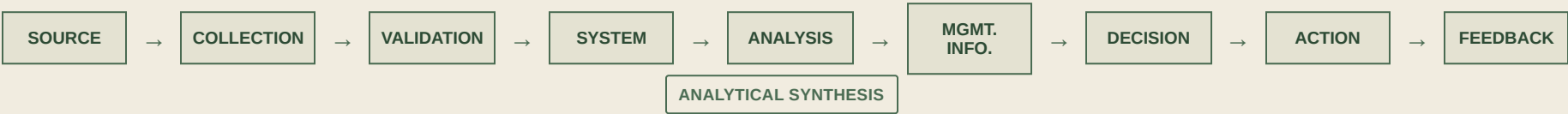
¹ Hammer & Champy, Reengineering the Corporation, 1993

² APQC, Process Classification Framework

DATA, TECHNOLOGY & INFORMATION FLOWS

What Information Does Management Need?

Information as an execution system: where it originates, how it is validated, and whether it reaches the **decision-maker** in usable form.



DATA ≠ INFORMATION ≠ MANAGEMENT INFORMATION ≠ REPORTING

HISTORICAL Ackoff's data → information → knowledge hierarchy (1989) is the standard academic anchor for the first distinction — data as raw facts, information as data given meaning through relationships.¹ This research pass confirmed the citation but not Ackoff's exact original wording; treat as paraphrase. "Management information" (structured for decision-making — a number attached to a threshold, an owner, and a next action) and "reporting" (communication of results, often after the decision window has closed) are D4's own working definitions extending this base. The practical difference: a report can be entirely accurate and still be management information's opposite, if it arrives after the decision it should have informed, or in a form the decision-maker cannot act on without further translation.

DATA OWNERSHIP, STEWARDSHIP & LINEAGE

ANALYTICAL SYNTHESIS Three distinct roles are frequently conflated. **Ownership** is accountability for a data domain's definition and quality standard. **Stewardship** is the operational work of maintaining that standard day to day, often held by a different person than the owner. **Lineage** is the traceable record of where a number originated and what transformed it before it reached a report — without lineage, two teams citing "the same" figure cannot establish whether they mean the same thing. Common definitions across functions (what exactly counts as an "active customer," a "completed project") are a prerequisite for all three; without them, ownership and stewardship are assigned to a term whose meaning still varies by function.

TECHNOLOGY AS ENABLER, NOT SOLUTION

ANALYTICAL ILLUSTRATION Automating a poorly designed process can simply make the bad process faster — a direct corollary of Hammer's reengineering argument (Page 10): redesign must precede automation, not follow it. A new system implemented on top of undefined ownership and inconsistent definitions inherits both problems rather than resolving them.

DATA QUALITY — CORRECTED

DAMA International's own official material does not itself list a specific set of data-quality dimensions.² Accuracy, completeness, consistency, timeliness, validity and uniqueness are common practitioner terms — **GENERAL PRACTICE / ANALYTICAL SYNTHESIS** not a verified DAMA-endorsed taxonomy. They remain useful as a working checklist for diagnosing why a specific number is untrustworthy, provided they are not cited as an official standard.

SYSTEM FRAGMENTATION

Disconnected systems and "multiple versions of truth" extend Kaplan & Norton's already-established process-fragmentation finding (Page 1/2) to the data layer: the same underlying process fragmentation that produces disconnected planning, budgeting and review cycles also produces disconnected systems of record, because each fragmented process tends to accumulate its own system over time.

THE CHAIN — D4 SYNTHESIS

ANALYTICAL SYNTHESIS Data quality → information quality → decision quality. Poor data quality does not stay contained at the source: it propagates through every downstream analysis and report built on it, so the cheapest place to fix a decision-quality problem is usually upstream, at the data layer, not downstream in the reporting layer.

THE OBJECTIVE

Not "collect more data" — ensure decision-relevant information reaches the right decision-maker in reliable, timely, usable form.

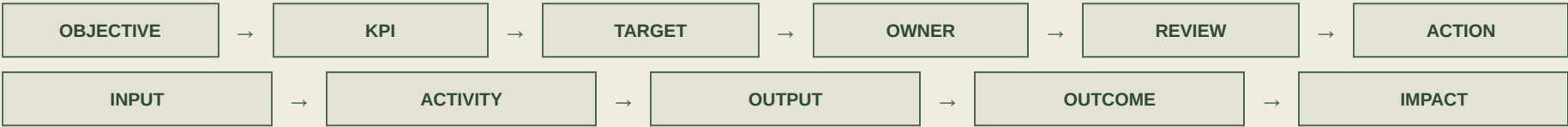
¹ Ackoff, "From Data to Wisdom," 1989

² DAMA International, DAMA-DMBOK, 2nd Ed. 2024

³ Kaplan & Norton, HBS WP 05-071

How Does Management Know It Is Working?

A dashboard can measure performance without changing it. Performance management requires measurement plus review, decision and **action**.



INSTITUTIONAL GUIDANCE

World Bank IEG: Input = "financial, human, and material resources"; Outcome = "short and medium-term effects... usually behavioural or organizational"; Impact = "positive and negative, primary and secondary long-term effects... intended or unintended."¹

RESULTS-MANAGEMENT "IMPACT"

The long-term effect of an organization's own initiative on its own intended outcomes (World Bank).

D2 "IMPACT MATERIALITY"

Severity, scale and scope of an organization's effects on external stakeholders (double materiality). A different literature, same word.

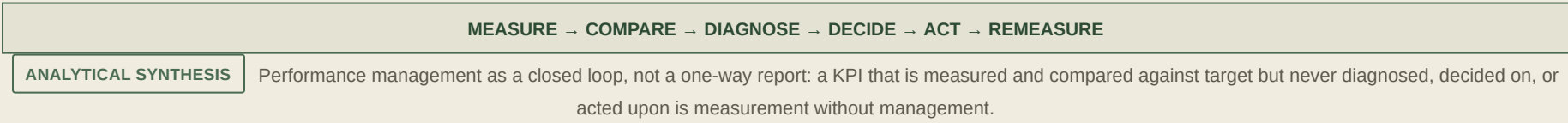
KPI VS. METRIC; LEADING VS. LAGGING

ANALYTICAL SYNTHESIS

A metric is any measured quantity; a KPI is a metric an organization has deliberately selected as a signal of progress against a stated objective — every KPI is a metric, not every metric is a KPI. A **lagging** indicator reports a result after the fact (revenue, churn); a **leading** indicator reports a condition believed to predict that result before it lands (pipeline coverage, employee engagement). A KPI architecture built only on lagging indicators tells management what already happened, too late to change it — the OBJECTIVE → KPI → TARGET chain above should include both.

ANALYTICAL SYNTHESIS

KPI cascades connect enterprise-level KPIs to business-unit and team-level KPIs beneath them; done well, a front-line KPI is visibly a contributor to the KPI above it. Cascading without that visible logic just multiplies the number of things measured, which is itself a form of the fragmentation problem from Page 6.



BALANCED SCORECARD

MANAGEMENT FRAMEWORK

Kaplan & Norton, 1992, original wording: Financial ("How do we look to shareholders?"), Customer ("How do customers see us?"), Internal Business ("What must we excel at?"), **Innovation and Learning** ("Can we continue to improve and create value?").² "Learning and Growth" is later (1996) terminology, not the 1992 original. One influential framework — not the universal KPI architecture; its contribution is the discipline of measuring across more than one perspective, not the specific four labels.

OWNERSHIP & REVIEW CADENCE

A KPI without a named owner is measured by no one in particular, which in practice means no one. Review cadence should match the KPI's own rate of change: a KPI that moves quarterly reviewed weekly wastes attention on noise; one that moves daily reviewed quarterly misses the window to act.

MEASUREMENT ≠ MANAGEMENT

A dashboard is measurement. Performance management additionally requires that someone compares the measurement to a target, diagnoses the gap, decides what to do, and is checked on whether they did it — the loop at left. Many organizations have extensive measurement and comparatively little management.

MEASURING THE WRONG THING

ACADEMIC EVIDENCE

Kerr (1975): "people focus on what is rewarded," even when the organization sincerely hopes for a different behaviour — his own title for this is "rewarding A while hoping for B." Vietnam-era rotation policy and individually-visible sports statistics both show reward systems producing behaviour opposite to organizational goals.³ A KPI architecture inherits this risk the moment any KPI is tied to reward — Page 14 develops the incentive-design implications.

¹ World Bank IEG, Glossary of Key Terms

² Kaplan & Norton, HBR 1992

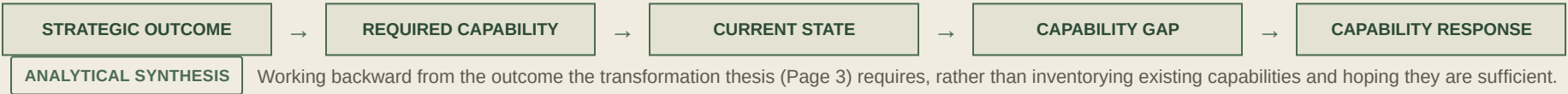
³ Kerr, AMJ 1975

CAPABILITIES & ORGANIZATIONAL DESIGN

What Must the Organization Be Able to Do?

Capability is **collective** and organizational — not an individual's skill, and not confined to a single function.

CONCEPT	DEFINITION
Capability	Organizational, collective ability. Prahalad & Hamel: "the collective learning in the organization... how to coordinate diverse production skills and integrate multiple streams of technologies." ¹ It survives any one individual leaving.
Skill	Individual proficiency — implied contrast, not separately sourced. A skill leaves the organization when the person does; a capability, properly built, does not.
Resource	An input or asset available to the organization — capital, headcount, a system, a dataset. A resource enables a capability but is not itself one: owning analytics software is a resource; being able to reliably turn data into a decision is the capability. ANALYTICAL SYNTHESIS
Function	An organizational unit (e.g. Finance, Marketing) with a defined scope and reporting line. ANALYTICAL SYNTHESIS
Capability vs. function	A capability can span multiple functions and business units — Prahalad & Hamel's tree metaphor: core competence as the root system feeding many branches, each branch a different product or business unit drawing on the same underlying strength. ¹



BUILD, BORROW, BUY — & PARTNER

Build
Develop the capability internally. Capron & Mitchell, *Build, Borrow, or Buy*, HBR Press 2012.² Slowest, most durable, highest control.

Borrow
Access it through alliance or licensing. Capron & Mitchell.² Faster, lower control, dependent on the partner's continuity.

Buy
Acquire an organization that already has it. Capron & Mitchell.² Fastest, most expensive, carries integration risk.

Partner
Ongoing joint capability with a third party. Not part of the original three-term framework. ANALYTICAL SYNTHESIS

CMMI — TREATED CAUTIOUSLY

INSTITUTIONAL

Originated to assess U.S. Department of Defense software contractors (SEI, Carnegie Mellon).³ CMMI Institute's own current material claims broad, cross-industry applicability, well beyond that original software-contractor scope — a vendor scope claim, not independent academic validation. Its five levels (Initial → Optimizing) should not be treated as a universally validated general enterprise-capability standard; they remain a defensible reference point specifically for process-maturity assessment, within that more limited claim.

CAPABILITY PLACEMENT

Centralized, embedded, center of excellence, or shared service — the same centralization/decentralization contingency logic from Page 7/8 applied to capability specifically. A centralized capability protects scarce expertise and consistency at the cost of responsiveness; embedding it in business units trades that consistency for speed and contextual fit; a shared-service model attempts a middle path — common infrastructure with some dedicated local contact — at the cost of added coordination overhead.

¹ Prahalad & Hamel, HBR 1990

² Capron & Mitchell, HBR Press 2012

³ CMMI Institute / SEI, Carnegie Mellon

What Actually Gets Rewarded and Funded?

A strategy with no **resource allocation** behind it is unlikely to be executed as intended.

WHAT THE STRATEGY SAYS

The formal, stated priority and its intended resourcing.

WHAT GETS FUNDED / REWARDED

Bower: the initiatives that receive capital and management attention reveal the organization's de facto priorities.¹

ACADEMIC EVIDENCE

Bower's field research on capital-investment decisions found actual strategy substantially determined by which proposals middle managers advance and which senior managers fund — independent of the formal strategy document.¹ Applied to the operating model specifically: a target operating model unbacked by corresponding budget, staffing and management attention remains unfunded intention, regardless of how it is described in a strategy deck.

ACADEMIC EVIDENCE

Kerr (1975, Page 12): a KPI that becomes a reward criterion inherits Kerr's documented failure modes — gaming, visible-metric fixation, unintended behaviour — if not carefully designed.² Incentive alignment means the reward structure and the strategic priority point the same direction; misalignment is not usually deliberate — it is what happens when a reward system, designed for a prior priority, is left running unchanged after the strategy changes (the same fragmentation mechanism from Page 2).

ANALYTICAL SYNTHESIS

A further common tension: short-term, easily measured results (this quarter's cost savings) are naturally easier to reward than long-term, harder-to-attribute results (a capability that pays off over several years) — which structurally biases incentive systems toward the former unless the reward design deliberately corrects for it.

STRATEGY → PRIORITIES → BUDGET/CAPITAL → INITIATIVES → CAPABILITY → PERFORMANCE → RE-ALLOCATION

ANALYTICAL SYNTHESIS

A closed loop, not a one-way allocation: performance evidence should feed back into the next allocation cycle, which is what the final arrow represents.

D1 FIREWALL

Capital-allocation *governance* (who decides, what criteria, what process) is D4's domain. Capital-allocation *valuation mechanics* — NPV, DCF, WACC, multiples — belong to D1 and are not addressed here.

ENTERPRISE VS. LOCAL OPTIMIZATION

A business unit optimizing its own KPI without integration mechanisms is the failure mode Lawrence & Lorsch's research documents (Page 6/8) — not a new claim. Locally rational funding decisions (each unit backs its own strongest proposal) can still produce an enterprise-level portfolio that is poorly diversified or misaligned with the overall thesis; resolving this is a portfolio-governance function (Page 16), not an individual-unit one.

¹ Bower, HBS 1970

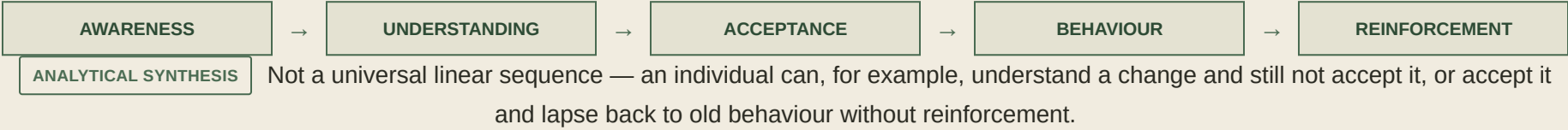
² Kerr, AMJ 1975

³ Lawrence & Lorsch, 1967

CHANGE MANAGEMENT

How Do People Adopt the New Way of Working?

Communication is not **adoption**. Popular change frameworks deserve the same source discipline applied everywhere else in this dossier.



Communication ≠ adoption. Communicating a change explains what is changing and why; adoption is the point at which people actually work differently, day to day, without being reminded. A well-communicated change with no reinforcement mechanism routinely produces understanding without behaviour change — people can accurately describe the new process while continuing to run the old one. **Project completion ≠ adoption** for the identical reason: the project team can disband having delivered every planned output while the target behaviour has not yet become the default.

RESISTANCE — ONE VERIFIED ACADEMIC ANCHOR

ACADEMIC EVIDENCE

Oreg (2003), *Journal of Applied Psychology*: a validated dispositional measure identifying four resistance mechanisms — routine seeking, emotional reaction to imposed change, short-term focus, cognitive rigidity.¹ Correlated with personality traits; unrelated to cognitive ability. Oreg studied **individual disposition**, explicitly not organizational or contextual causes of resistance — those (incompatible incentives, accountability without authority) are addressed via Kerr and Rogers & Blenko elsewhere in this dossier. Practically, this means resistance is not always a signal to persuade harder: some resistance traces to a genuinely misaligned incentive or an unresolved decision-rights conflict, which Oreg's framework does not diagnose and a communication campaign will not fix.

CHANGE FATIGUE

ANALYTICAL ILLUSTRATION

Repeated, poorly sequenced or poorly reinforced change initiatives can exhaust an organization's capacity to absorb further change, independent of any single initiative's merits — a reason sequencing (Page 17) and embedding (Page 19) are treated as part of change management, not separate concerns.

LEWIN — CORRECTED

ACADEMIC EVIDENCE

Cummings, Bridgman & Brown (2016), *Human Relations*: Lewin never wrote "refreezing"; "unfreeze—change—refreeze" appears as 131 words in a 338-page book, later popularized by his students and business-school textbooks.² Not uncontested Lewin doctrine — treat the three-word cycle as a widely-taught simplification of Lewin's actual, considerably richer field-theory work, not as his own headline formulation.

ADKAR

CONSULTING FRAMEWORK

Prosci / Jeff Hiatt: Awareness, Desire, Knowledge, Ability, Reinforcement — a sequential, individual-level model of how one person moves through a change. Prosci's own benchmarking claims (e.g. "seven times more likely") are vendor claims, not independently verified here, and are not repeated as fact in this dossier.

LEADERSHIP BEHAVIOUR ≠ COMMUNICATION

What leaders actually fund and approve (Bower, Page 2/14) — not what they say in a town hall — is the evidenced form of sponsorship. Employees calibrate to observed leadership behaviour (what gets asked about, funded, and tolerated) considerably faster than to stated messaging, which is why a change with strong communications but unchanged leadership behaviour tends to be read as not genuinely mandated.

REINFORCEMENT

The mechanism that keeps a new behaviour from decaying back to the old default once attention moves elsewhere — typically some combination of updated performance review criteria (Page 12), updated incentives (Page 14), and continued visible leadership behaviour, not a single one-time reinforcement event.

1

Oreg, JAP 2003

2

Cummings, Bridgman & Brown, Human Relations 2016

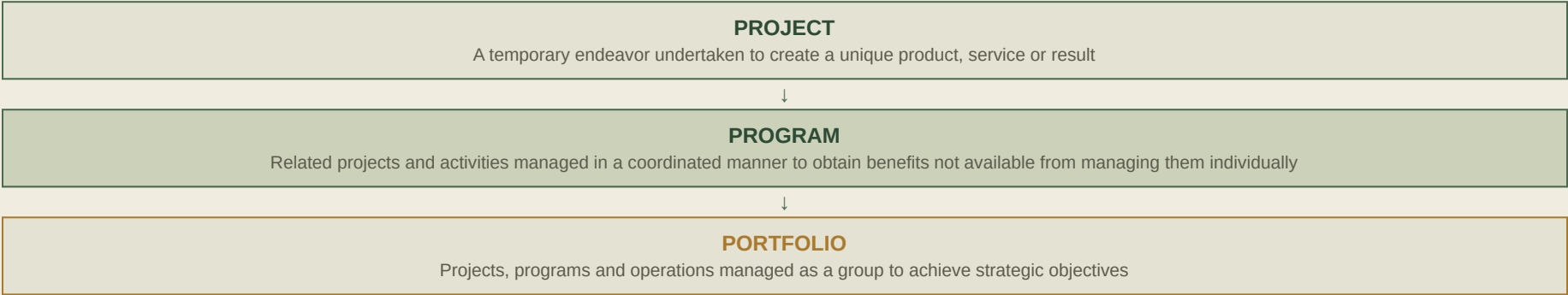
3

Prosci / Hiatt, ADKAR

THE TRANSFORMATION PORTFOLIO

Doing the Work Right vs. Doing the Right Work

A project delivered on time and on budget is not **evidence** the organization funded the right project in the first place.



INSTITUTIONAL GUIDANCE

 Definitions per PMI's standards for project, program and portfolio management.¹

TWO DIFFERENT QUESTIONS

PROJECT / PROGRAM LEVEL

Is this specific piece of work being executed well — on scope, on schedule, on budget, to quality? A discipline of delivery.

PORTFOLIO LEVEL

Is this the correct set of initiatives, given strategy and constrained capital and capacity, out of every initiative the organization could have funded instead? A discipline of selection.

ANALYTICAL SYNTHESIS

 "Doing the work right" (project/program discipline) versus "doing the right work" (portfolio discipline) is a D4 formulation of this distinction, not PMI's own phrasing — the two questions are separable and a portfolio can fail even when every individual project inside it succeeds.

WHAT CHANGES AT EACH LEVEL

LEVEL	TIME HORIZON	PRIMARY SUCCESS MEASURE
Project	Bounded, defined start and end	Scope, schedule, budget, quality
Program	Coordinated set of related projects	Benefits not available from managing them separately (Page 18)
Portfolio	Ongoing, cyclical, tied to strategic planning	Achievement of strategic objectives; correctness of the initiative mix itself

ANALYTICAL SYNTHESIS

 Table structure is a D4 reading of the three PMI definitions above, not a PMI-published comparison table.

WHAT PORTFOLIO MANAGEMENT ACTUALLY INVOLVES

ANALYTICAL SYNTHESIS

Prioritization ranks candidates against the transformation thesis (Page 3), not sponsor conviction. **Balancing** checks the mix as a whole — competing for the same scarce capability (Page 13) or sharing the same risk defeats the point of a portfolio. **Dependencies** map which initiatives require another first (Page 17). **Resource and capacity** checks confirm the organization has enough people and attention for the portfolio collectively — justified initiative-by-initiative can still exceed what the organization can deliver, **portfolio overload**.

KEY TAKEAWAY

A perfectly executed project inside a wrongly composed portfolio still fails the transformation.

A WELL-RUN PROJECT, A FAILED PORTFOLIO

A project executed flawlessly against the wrong priority is a portfolio failure, not a project-management failure. Bower's finding (Page 2/14) — that funded initiatives reveal an organization's real strategy — is precisely a portfolio-composition question, not a delivery question.

PORTFOLIO GOVERNANCE

Which forum decides what enters or exits the portfolio, on what criteria, at what cadence — the Page 9 governance/decision-rights logic applied specifically to initiative selection and capital allocation (Page 14). Entry and exit criteria matter equally: a portfolio that only ever adds initiatives, without a mechanism for retiring or pausing lower-priority ones, is the most common route into overload.

D1/D2/D3 FIREWALL

Portfolio prioritization criteria may draw on financial materiality (D1), impact materiality (D2) and climate-scenario stress-testing (D3) as inputs. This dossier does not re-derive those valuation, stakeholder-impact or scenario methodologies — only the governance mechanism that decides among initiatives once those inputs exist.

¹ PMI, The Standard for Portfolio Management

² PMI, A Guide to the Project Management Body of Knowledge (PMBOK Guide)

TRANSFORMATION ROADMAP & SEQUENCING

In What Order Should Change Happen?

Sequencing logic borrowed carefully from project **scheduling** — not itself a transformation methodology.

ROADMAP VS. PROJECT PLAN

ANALYTICAL SYNTHESIS

A project plan sequences the tasks inside one bounded piece of work, typically owned by one project manager. A roadmap sequences multiple initiatives across the whole transformation portfolio (Page 16) against shared dependencies and shared capacity — it operates one level up, and is owned at the portfolio-governance level, not by any single project's manager.

CRITICAL PATH METHOD — WHAT IT ACTUALLY IS

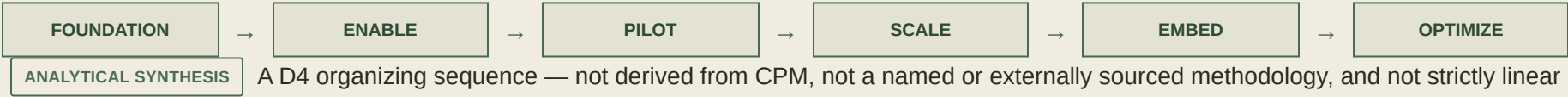
HISTORICAL

The Critical Path Method was developed in the late 1950s by Morgan R. Walker (DuPont) and James E. Kelley Jr. (Remington Rand), published as "Critical-Path Planning and Scheduling" at the 1959 Eastern Joint Computer Conference, originally applied to industrial plant construction and maintenance scheduling.¹ CPM identifies the longest chain of dependent tasks, which sets the minimum possible duration of a project — a project-scheduling technique, not a change-management or transformation methodology.

ANALYTICAL ILLUSTRATION

D4 borrows only CPM's underlying idea — that dependencies, not preference or urgency, should determine sequence — as an analogy for transformation sequencing. This is not a claim that transformation programs are literally CPM-scheduled.

A D4 SEQUENCING LOGIC



ANALYTICAL SYNTHESIS

A D4 organizing sequence — not derived from CPM, not a named or externally sourced methodology, and not strictly linear in practice (see Page 20). **Pilot** tests the design against one bounded population before wider commitment; **scale** extends a validated pilot in planned waves rather than everywhere at once, so a design flaw surfaces while the cost of fixing it is still small; **wave planning** is the specific sequencing of which units or geographies receive the change in which order, typically prioritized by readiness and risk rather than convenience.

TWO SEQUENCING PHILOSOPHIES

FOUNDATION-FIRST

Resolve governance, decision rights and data foundations before scaling.
Slower visible progress; fewer prerequisite failures later.

QUICK-WINS-FIRST

Demonstrate early, visible results to build momentum and sponsorship.
Faster credibility; higher risk of rework if a later-addressed dependency invalidates early work.

ANALYTICAL ILLUSTRATION

Neither philosophy is independently evidenced as superior in this research; the trade-off follows directly from the dependency logic above, not from a separate study of sequencing strategy.

DEPENDENCY, NOT APPETITE

Governance and decision rights (Page 9) logically precede portfolio prioritization (Page 16) — an organization cannot rationally select initiatives before it knows who has authority to select them. Data and KPI foundations (Pages 11–12) logically precede performance-linked incentive redesign (Page 14) — incentives cannot be tied to measures that do not yet exist reliably.

A COMMON SEQUENCING FAILURE

ANALYTICAL ILLUSTRATION

Sequencing driven by organizational appetite, politics or visibility — rather than genuine dependency — produces work that must later be redone once a prerequisite is finally addressed.

PHASE GATES

A checkpoint at which the organization deliberately confirms a prerequisite phase is genuinely complete before releasing capital or attention to the next — the sequencing discipline made procedural, tied back to the portfolio governance forum (Page 9, Page 16).

¹ Kelley & Walker, "Critical-Path Planning and Scheduling," Proceedings of the Eastern Joint Computer Conference, 1959

TRANSFORMATION CONTROL TOWER

Deliverable, Capability, Benefit, Outcome — Not the Same Thing

A completed deliverable is not a realized **benefit**. A green status report can still sit on top of a failing transformation.



ANALYTICAL SYNTHESIS A four-term chain built on PMI's benefits-realization distinction, not PMI's own exact four terms. **Deliverable**: the completed output of a project. **Capability**: the durable organizational ability that deliverable enables (Page 13). **Benefit**: the measurable advantage realized once that capability is actually used. **Business outcome**: the strategic-level result the benefit contributes to (Page 3's transformation thesis).

INSTITUTIONAL GUIDANCE PMI's Benefits Realization Management practice guide holds that delivering a project's outputs is not the same as realizing its intended benefits: benefits typically continue to accrue, and must continue to be actively managed, after a project itself has closed.¹ A status report that stops at "deliverable complete" answers only the first link in this chain.



INSTITUTIONAL GUIDANCE PMI's own three-phase benefits-realization structure, shown here directly rather than folded into the four-term chain above — sustainment is PMI's explicit acknowledgment that benefits require ongoing management after execution ends, not a one-time handover.¹

THE CONTROL-TOWER IDEA

ANALYTICAL ILLUSTRATION "Control tower" here names a D4 concept — a portfolio-level visibility mechanism that tracks initiatives against all four terms above, not schedule and budget status alone. It is a D4 operationalization of the benefits-realization logic, not a named PMI or industry-standard artifact. A working control tower typically tracks, across the whole portfolio: **milestones** (is each initiative on its planned sequence), **dependencies** (is a prerequisite from another initiative at risk), **issues** (what is blocked and who owns unblocking it), **benefits** (is the realized benefit tracking to plan, not just the deliverable), **resources** (is capacity actually available where the roadmap assumes it), and **decisions** awaiting executive action — the point of consolidating all six in one place is that a problem visible only in isolation (a resourcing gap on one initiative) often explains a symptom visible elsewhere (a milestone slipping on a dependent initiative).

WORKING THROUGH THE CHAIN — ANALYTICAL ILLUSTRATION

TERM	ILLUSTRATION
Deliverable	A new customer-data system goes live
Capability	Staff can now see a unified customer record across business lines
Benefit	Time to resolve a customer query falls measurably
Business outcome	Retention or cross-sell improves at the level the transformation thesis (Page 3) targeted

ANALYTICAL ILLUSTRATION Constructed illustration, not a documented case — consistent with the illustrative convention used on Page 3.

THE RAG-STATUS TRAP

A project can be reported "green" on schedule, scope and budget while the capability it was meant to build is never actually used, or the benefit never actually materializes. Schedule/budget status and benefits-realization status are two different questions, answered by two different tracking mechanisms.

OWNERSHIP BEYOND PROJECT CLOSE

If no one owns the capability-to-benefit link after the project team disbands, the chain above breaks at exactly the point PMI's guidance identifies as most at risk.

PMO VS. TRANSFORMATION OFFICE

EVIDENCE GAP / GENERAL PRACTICE Practitioner usage varies: some organizations use "PMO" for a project-delivery support function and reserve "transformation office" for a broader, benefits-and-portfolio-level function closer to the control tower described here; others use the terms interchangeably. No single universal definition is asserted in this dossier — what the function does (tracks the six items at left, across the whole portfolio) matters more than which of the two names it carries.

RAG AS COMMUNICATION, NOT DIAGNOSIS

A red/amber/green status is a compact way to communicate a summary judgment upward; it is not itself an analysis of why a status is what it is. Treating RAG as diagnosis — debating the color rather than the underlying milestone, dependency or benefit-realization issue behind it — is a common way a control tower degrades into a reporting ritual.

¹ PMI, Benefits Realization Management: A Practice Guide, 2019

EMBEDDING & CONTINUOUS IMPROVEMENT

How Does the Transformation Stop Being a Project?

Deming's own preferred formulation was **Plan–Do–Study–Act** — not the "Check" that later popularizations substituted.



PRIMARY SOURCE The cycle originates with Walter Shewhart; Deming adapted and consistently taught it as Plan–Do–**Study**–Act (PDSA), deliberately preferring "Study" over "Check" because it emphasizes learning and building knowledge from the comparison of results to prediction, not mere verification.¹ "Plan–Do–Check–Act" (PDCA) became the more widely popularized formulation through others, including later Japanese quality-management practice; D4 does not attribute the PDCA wording to Deming himself.

INSTITUTIONALIZATION — BORROWED CAREFULLY

ACADEMIC EVIDENCE Tolbert & Zucker's classic study of civil service reform diffusion across U.S. state governments describes institutionalization in three stages: **habitualization** (a new practice is adopted and repeated, but only where formally mandated), **objectification** (the practice develops a shared meaning and normative consensus that it is the correct way to operate), and **sedimentation** (the practice persists across generations of people and survives even when its original champions have left).³ **ANALYTICAL INTERPRETATION** Their original study concerns how a practice diffuses *across many separate organizations* over decades; D4 applies the same three-stage logic *within one organization* embedding one transformation — an analogy for the shape of the process, not a claim that Tolbert & Zucker studied internal transformation embedding.

EMBEDDING = PERMANENT INFRASTRUCTURE, NOT A CLOSING TASK

Kaplan & Norton's argument for a small, *permanent* function (Page 4) applies directly here: a transformation stops being a temporary program only when a standing owner, cadence and mechanism exist to keep running the Plan–Do–Study–Act loop against the operating model itself, after the original transformation program has formally closed. Embedding specifically requires an explicit **ownership transfer** — from the transformation program team to the standing part of the organization that will run the new way of working permanently — and a matching **capability transfer**, so the receiving team can actually operate what was designed without depending on the original program team's tacit knowledge.

WHAT THE LOOP APPLIES TO

The same Plan–Do–Study–Act logic can run at two different grains: applied to a single process, it is ordinary operational improvement; applied to the operating model itself — its governance, its KPI architecture, its incentive design — it is what keeps the five systems on Page 20 from calcifying into the same rigid design that prompted the transformation in the first place. **ANALYTICAL SYNTHESIS**

TRANSFORMATION DECAY

ANALYTICAL ILLUSTRATION Without reinforcement (Page 15), a newly adopted behaviour can regress toward the prior default once program attention moves elsewhere — described here descriptively, not as a formally established academic theory. Behavioural reinforcement, a functioning feedback loop (measure → study → adjust), and a genuine program close-out that hands off ownership rather than simply stopping are the three factors that most directly determine whether decay occurs.

A MATURITY PROGRESSION — D4'S OWN CONSTRUCT

ANALYTICAL SYNTHESIS Ad Hoc → Defined → Managed → Continuously Improving. This progression is a D4 synthesis for describing how embedded a way of working has become; it is **not** CMMI's five-level model (Page 13, already treated cautiously) and should not be presented as derived from it.

WHO OWNS THE LOOP

Consistent with Page 4 and Page 18: a loop with no named, standing owner tends to stop running once the original program team disbands, even where the mechanism itself was well designed.

SCOPE NOTE

EVIDENCE GAP Kaizen and related continuous-improvement traditions were not sufficiently researched in this dossier and are deliberately not added here for breadth.

¹ The W. Edwards Deming Institute, on Shewhart/PDSA

² Kaplan & Norton, HBS WP 05-071

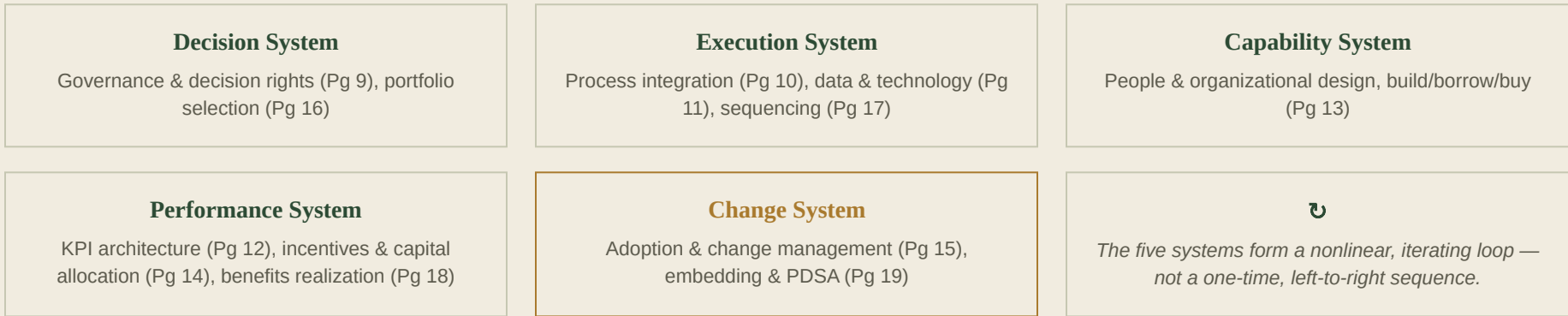
³ Tolbert & Zucker, "Institutional Sources of Change in the Formal Structure of Organizations," *Administrative Science Quarterly*, 1983

Strategy Becomes Execution Through Five Systems, Not One Event

Strategy defines direction. Operating model defines how the organization executes that direction. Transformation is the process of moving from one state to the other — and it is **never finished**.

ANALYTICAL SYNTHESIS — DOSSIER 4 MASTER FRAMEWORK

Assembled from Pages 1–19 of this dossier. Not an official or externally published framework — a D4 organizing device.



The eight dimensions introduced on Page 6 are grouped here into five interacting systems for synthesis purposes; the grouping compresses, but does not replace, that earlier detail. A weakness in any one system degrades the others — a decision-rights failure (Page 9) starves the performance system of enforceable accountability; a capability gap (Page 13) leaves the execution system unable to act on good decisions; a change-management failure (Page 15) prevents an otherwise sound design from ever being adopted in practice.

Transformation is the process of moving from one state to the other — and it is never finished.

D4 FINAL DIAGNOSTIC

Of any organization's transformation effort, ask, system by system:

- Decision** — who has the authority to decide?
- Execution** — does the work reach the customer or objective end-to-end?
- Capability** — can the organization repeat this without depending on specific individuals?
- Performance** — does measurement and reward point at the actual priority?
- Change** — will the new way of working survive the program team disbanding?

THE FOUR-DOSSIER BRIDGE

D1 (Matter) asks what is financially material. **D2 (Impact)** asks what the organization materially affects. **D3 (Future)** asks what could happen under different climate and transition paths. **D4 (Transformation)** asks whether the organization is actually capable of acting on any of it. The first three dossiers diagnose what matters and what might happen; this dossier diagnoses whether management can execute a response.

CLOSING SYNTHESIS

A strategy can be correct and still fail. Whether it does depends on the operating model built to execute it — and that model is a standing system to be governed, not a project to be closed.

1 This page synthesizes Pages 1–19 of this dossier; no new external sources are introduced.